

SPL-3 FINAL PRESENTATION



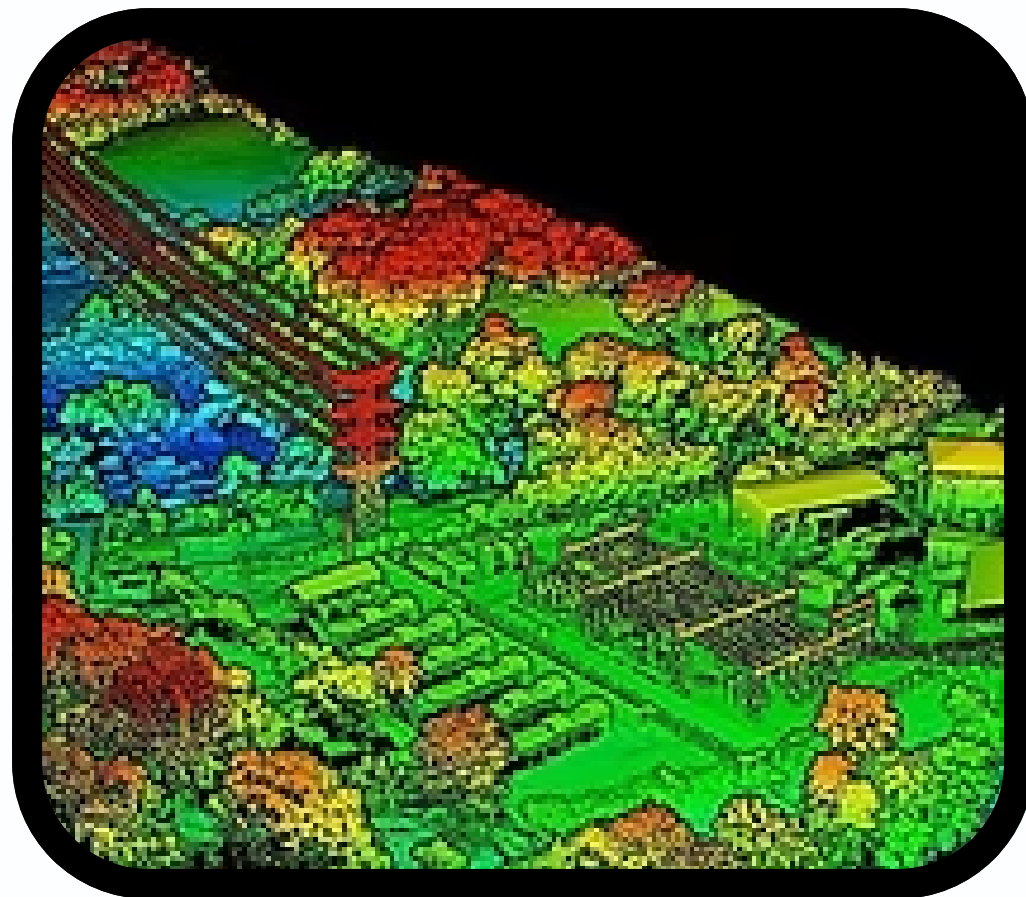
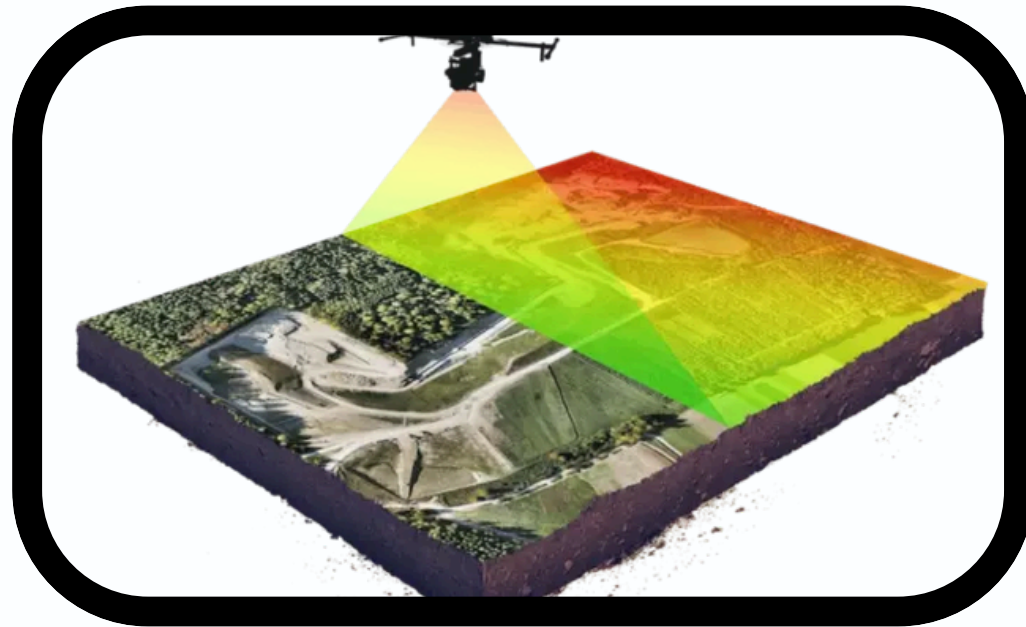
**BandContour: A Web-
Based System for LiDAR
Roof Contour Extraction**



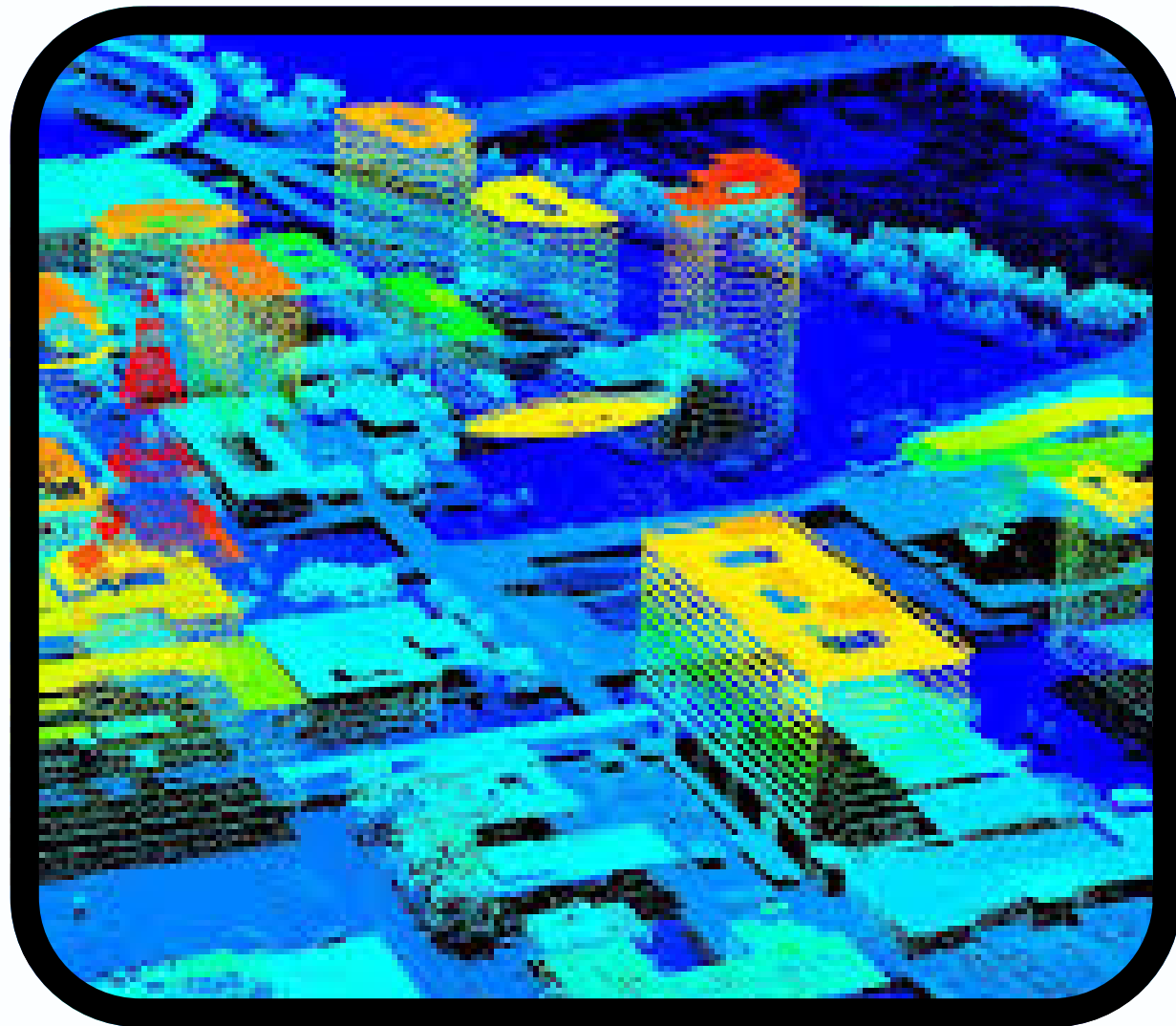
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Introduction > What is Point Cloud Data ?

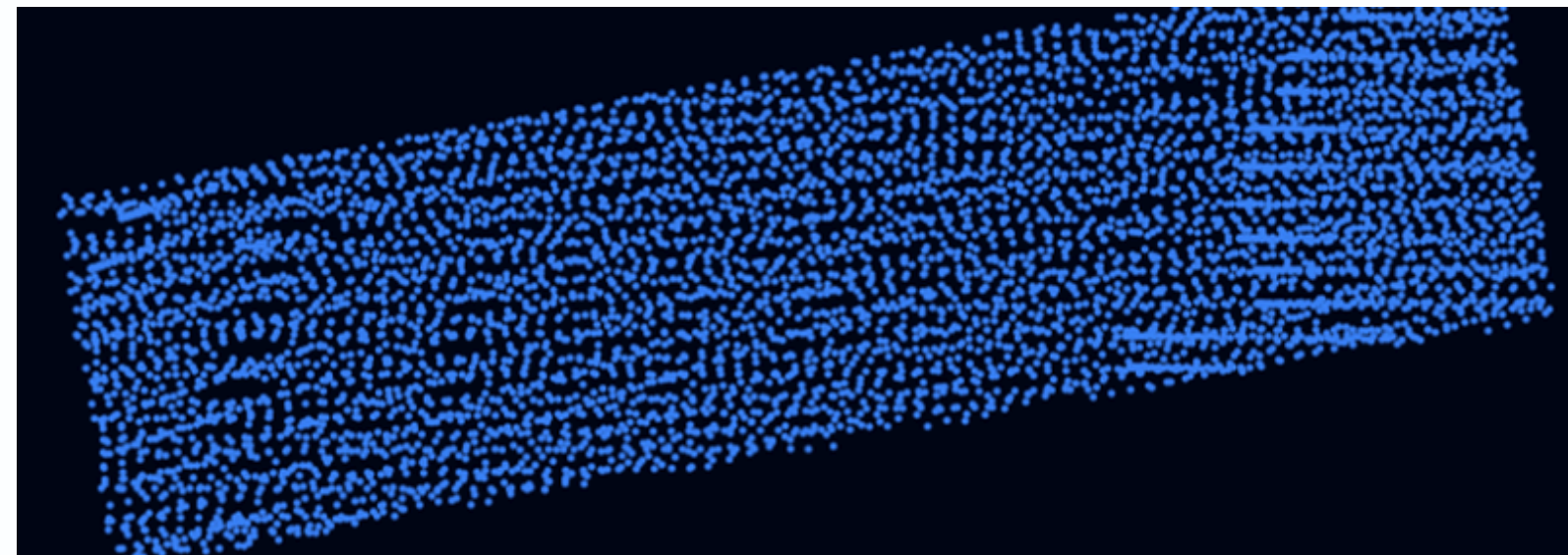
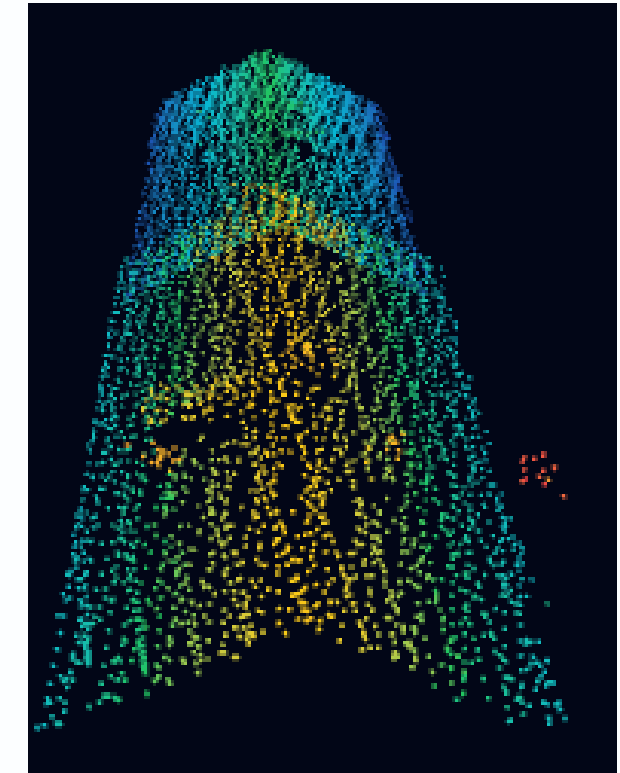
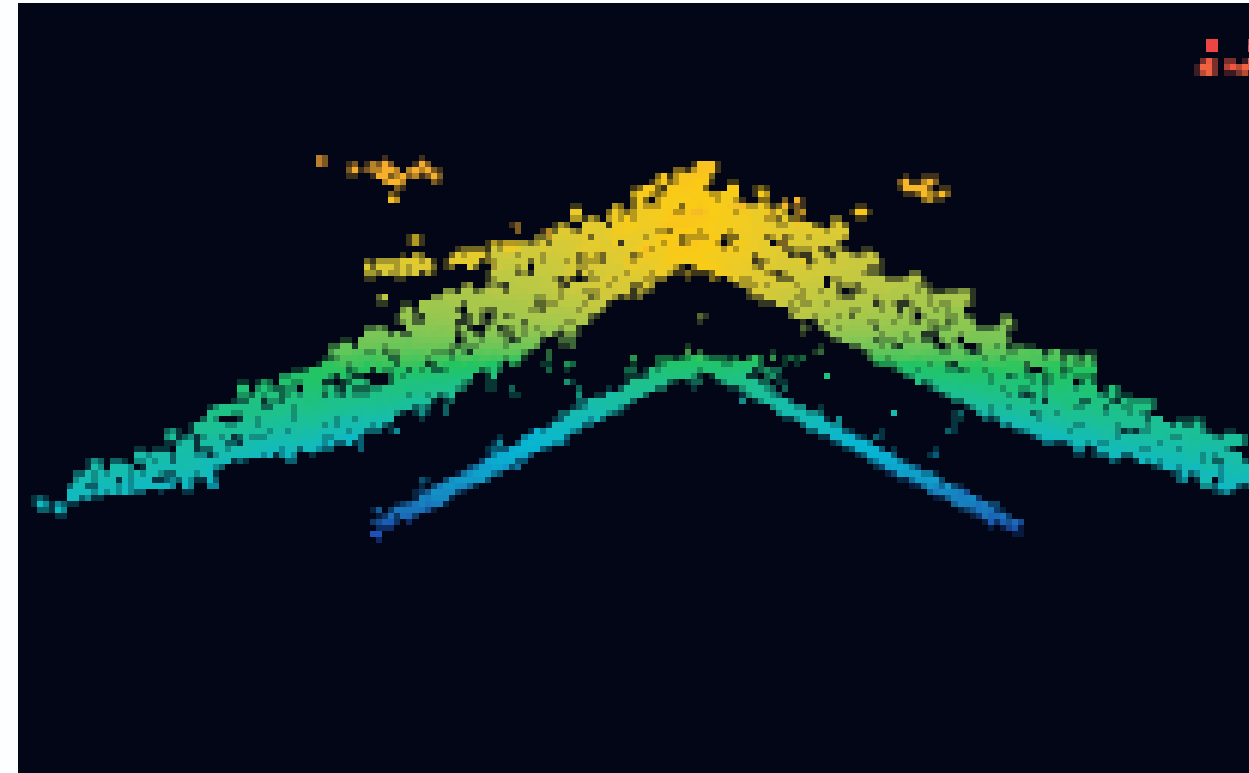
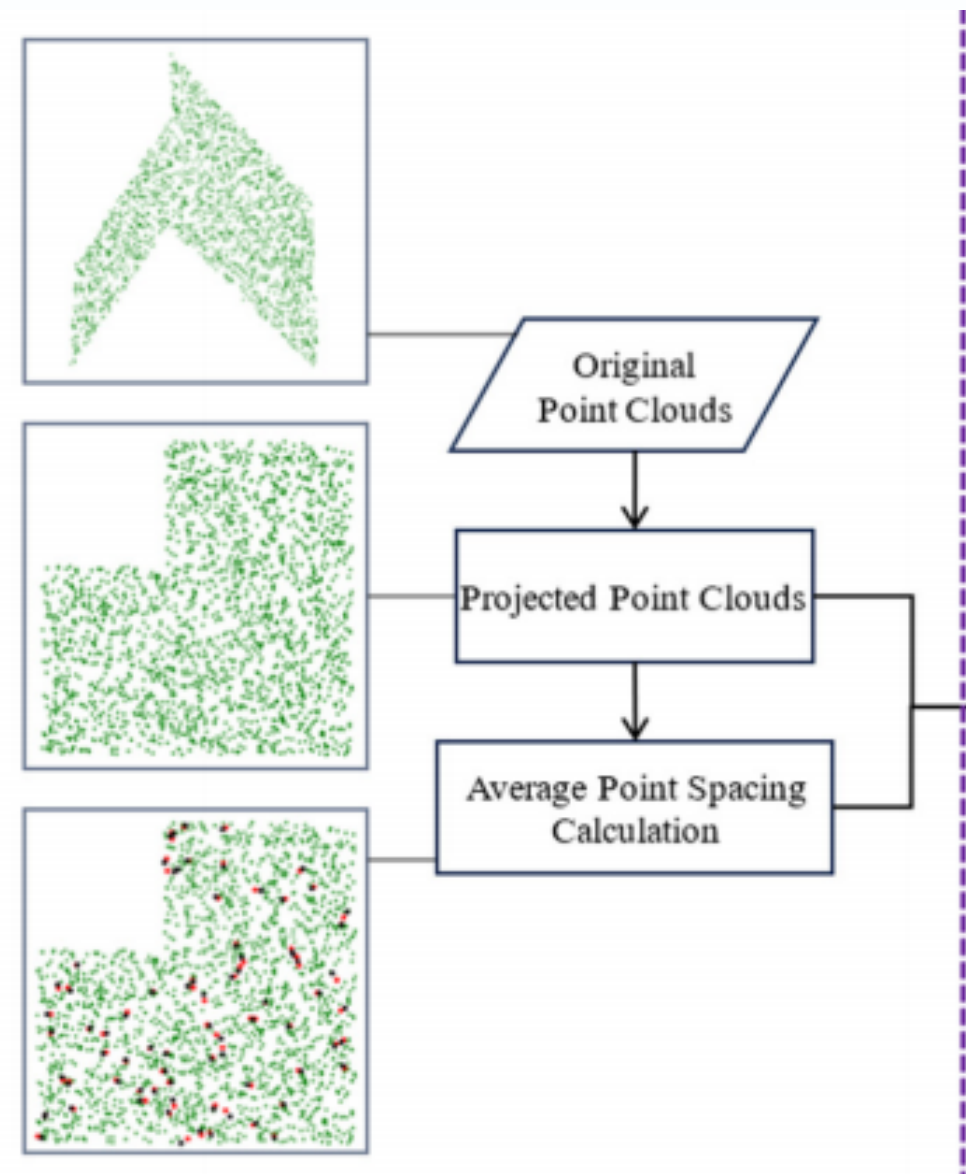


- A collection of millions of 3D points captured by LiDAR sensors.
- When we put all the points together, we can see the full shape of buildings, roads, and trees.

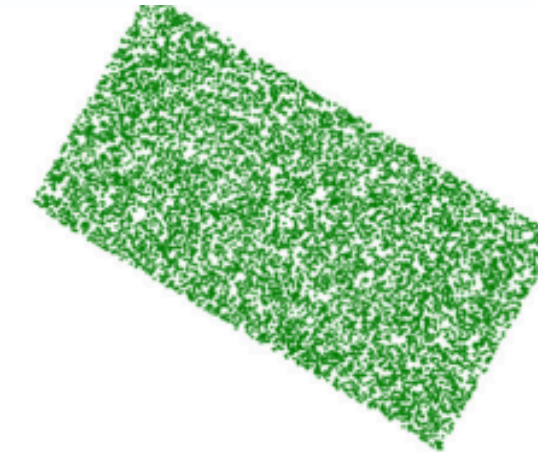
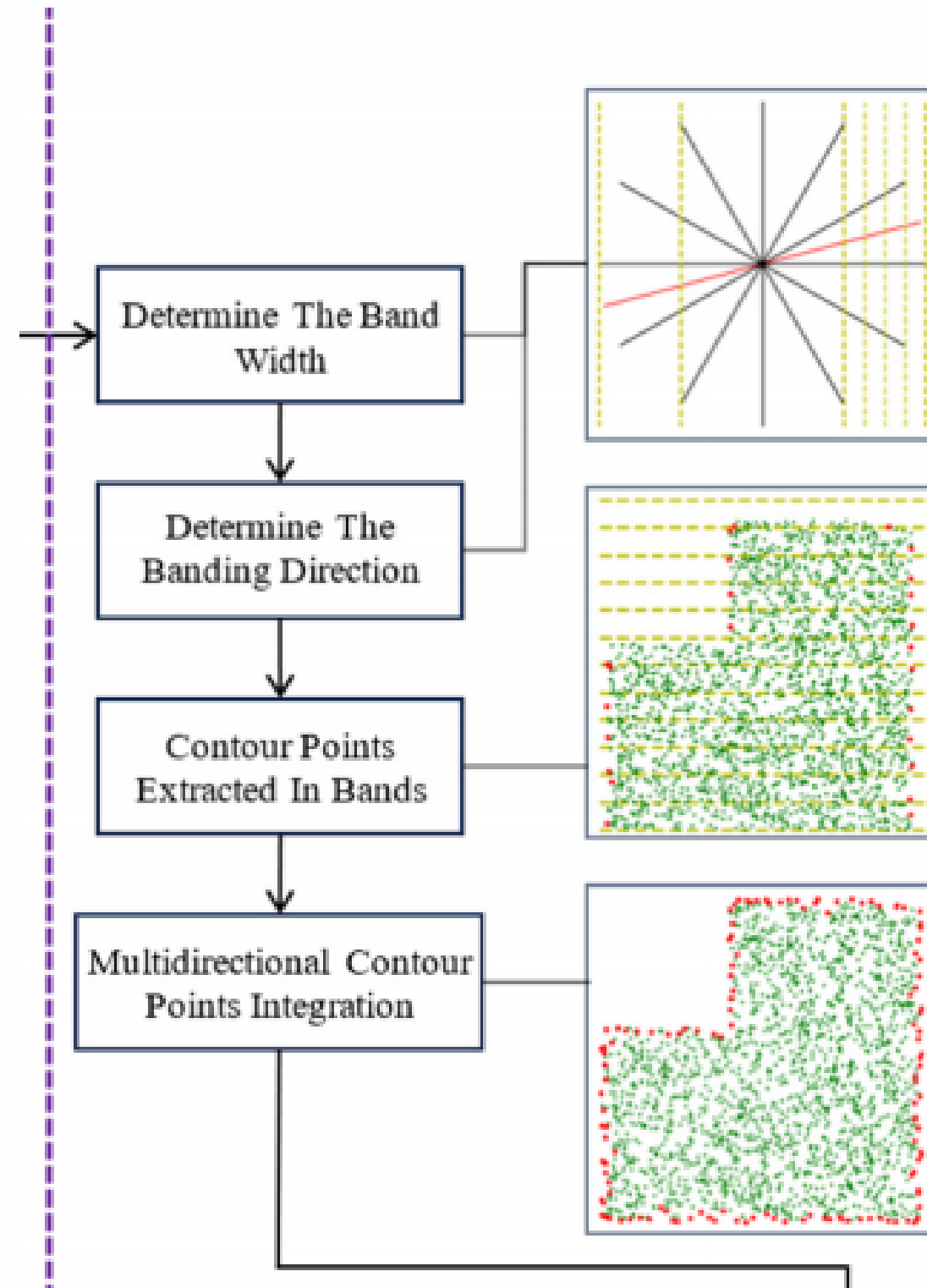


- Buildings are key elements in city modeling, urban planning, disaster management.
- Accurate building outlines are needed for planning, mapping, and 3D reconstruction.
- Manual extraction of building outlines is slow and error-prone. Automatic extraction improves speed, accuracy, and scalability.
- The project aims to develop a web-based system that can automatically detect and draw the outlines of building roofs.

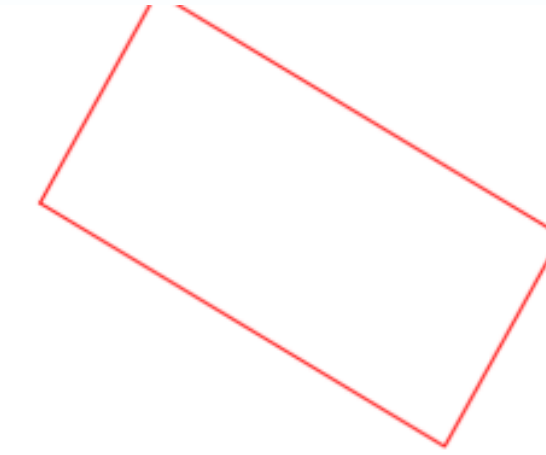
About Proposed Method (Preprocessing)



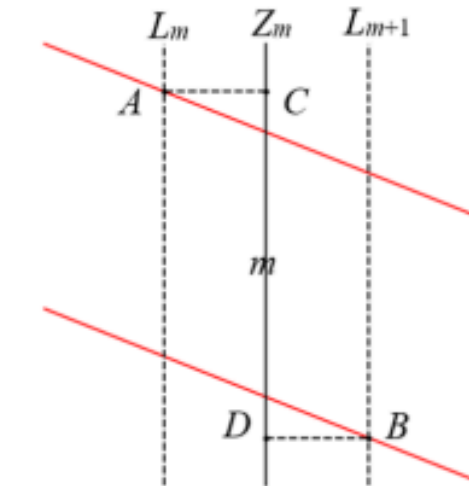
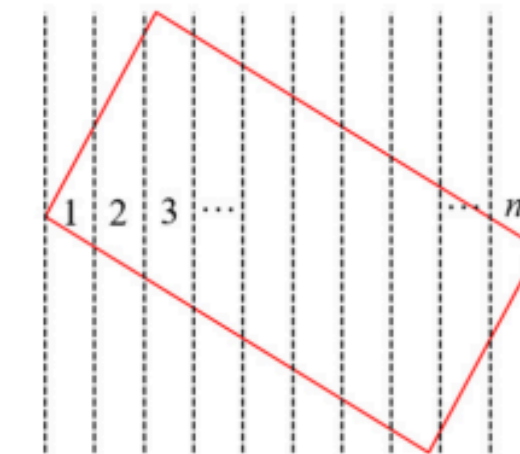
About Proposed Method (Contour Points Extraction - Single Direction)



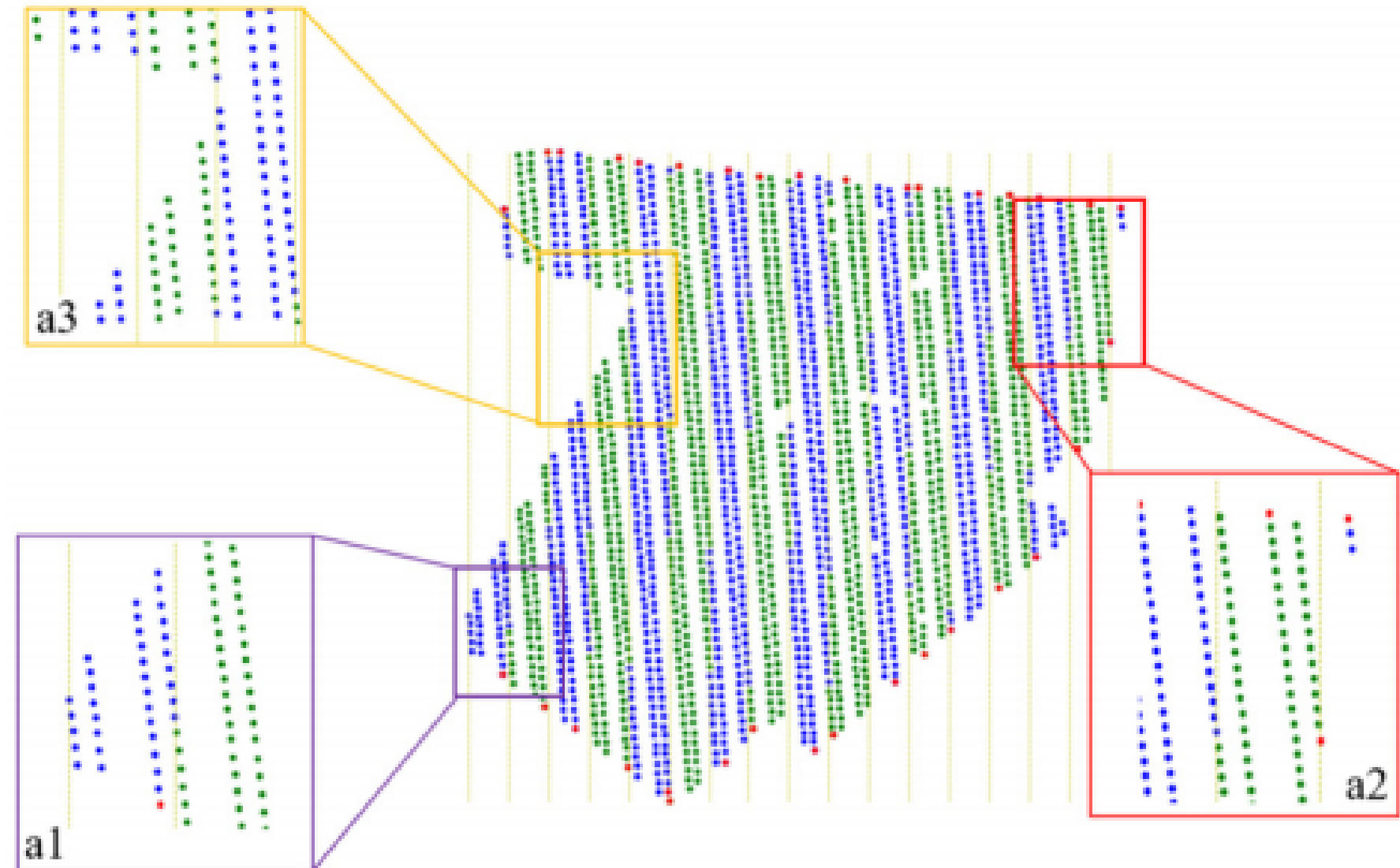
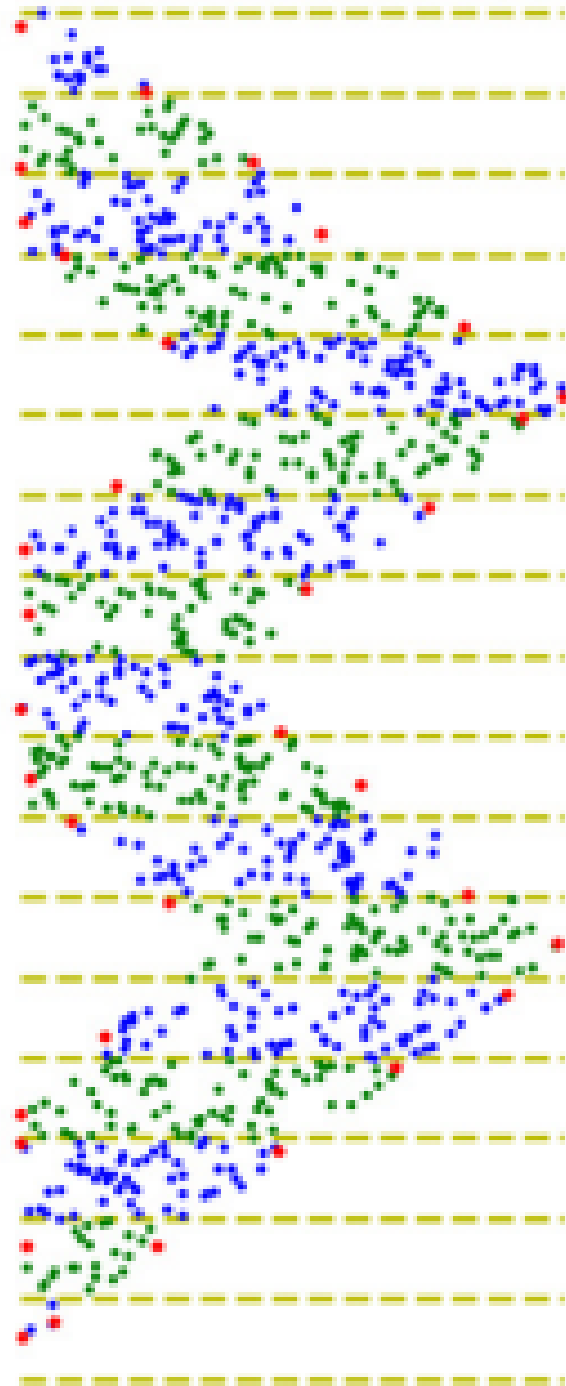
(a)



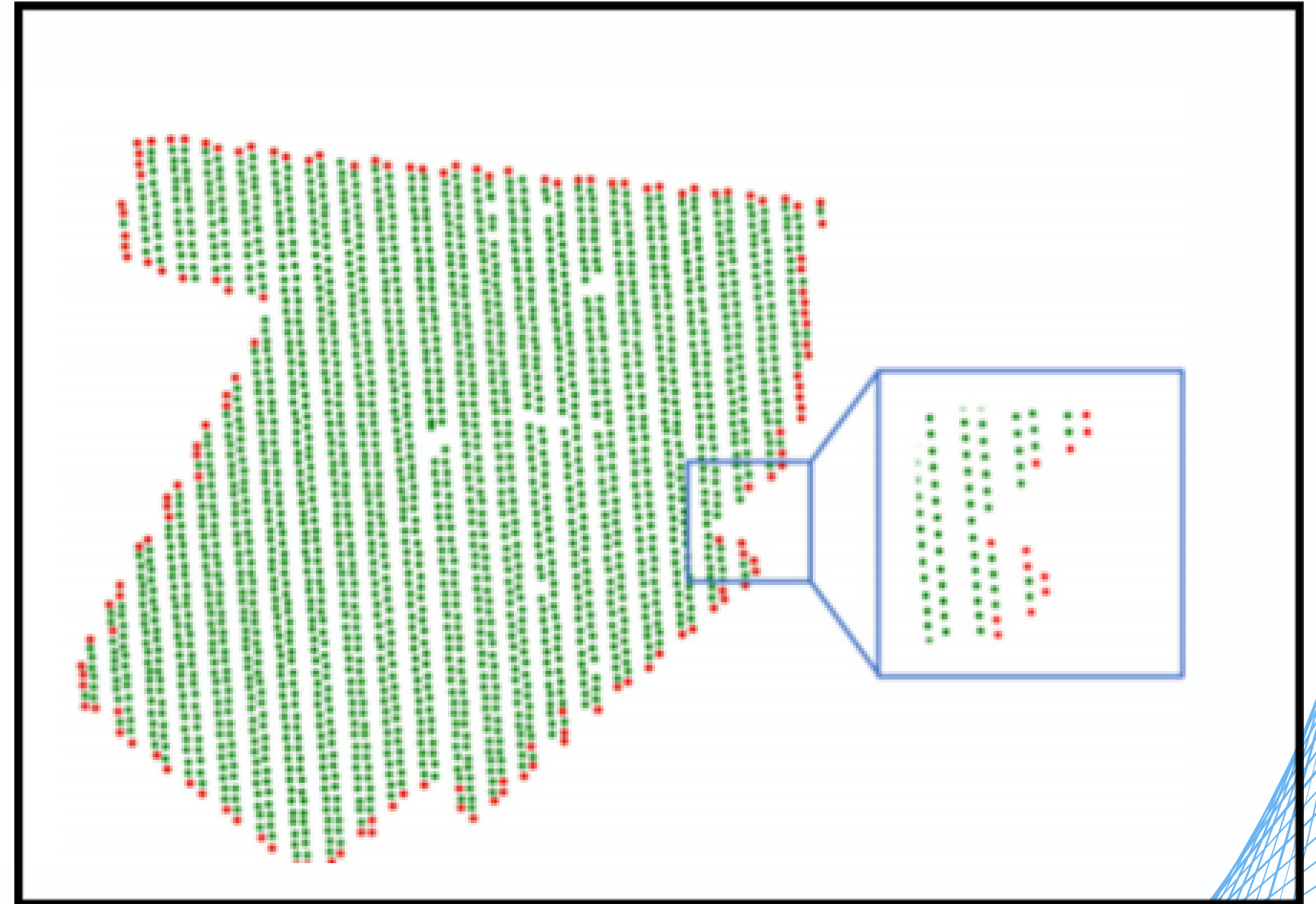
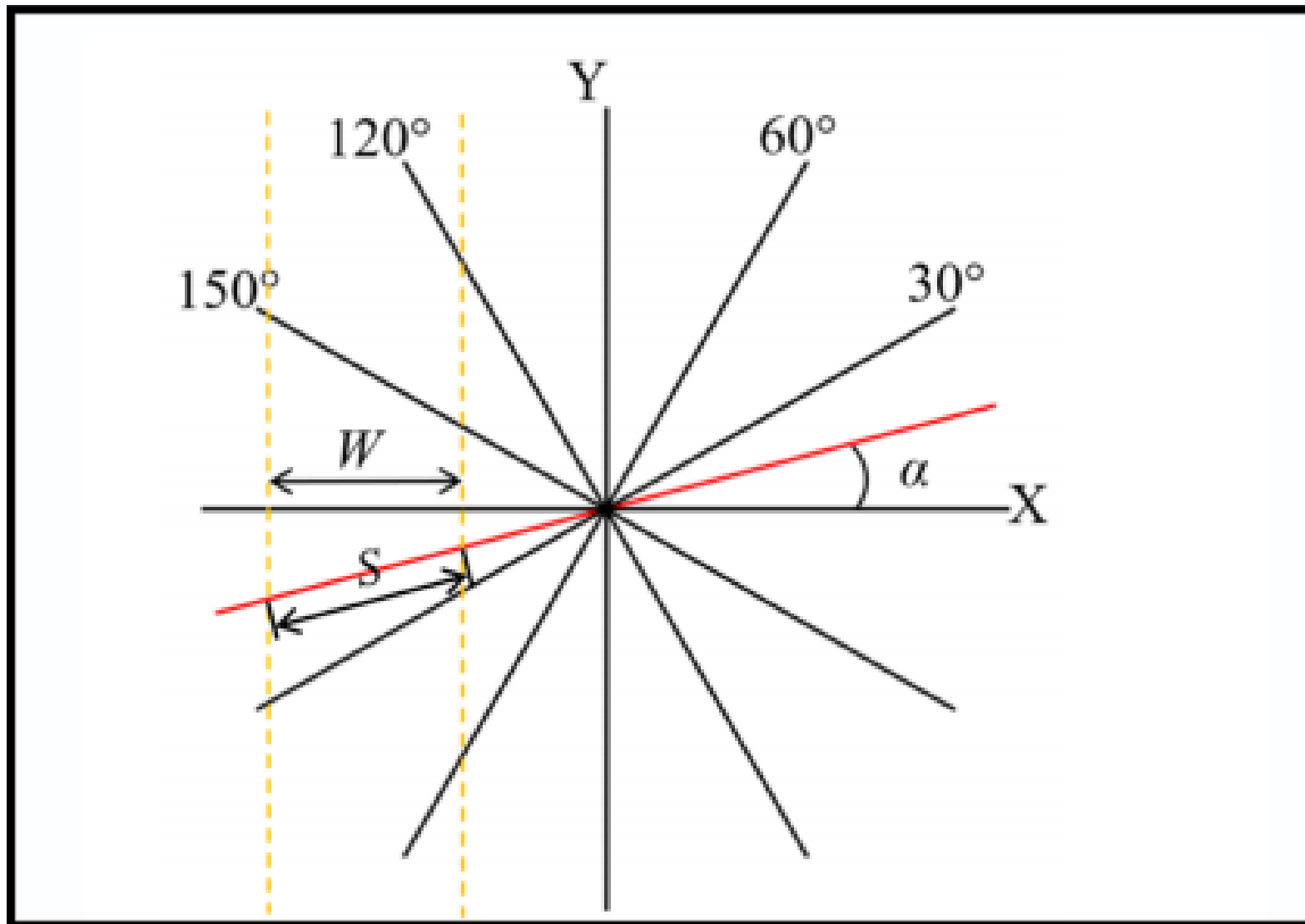
(b)



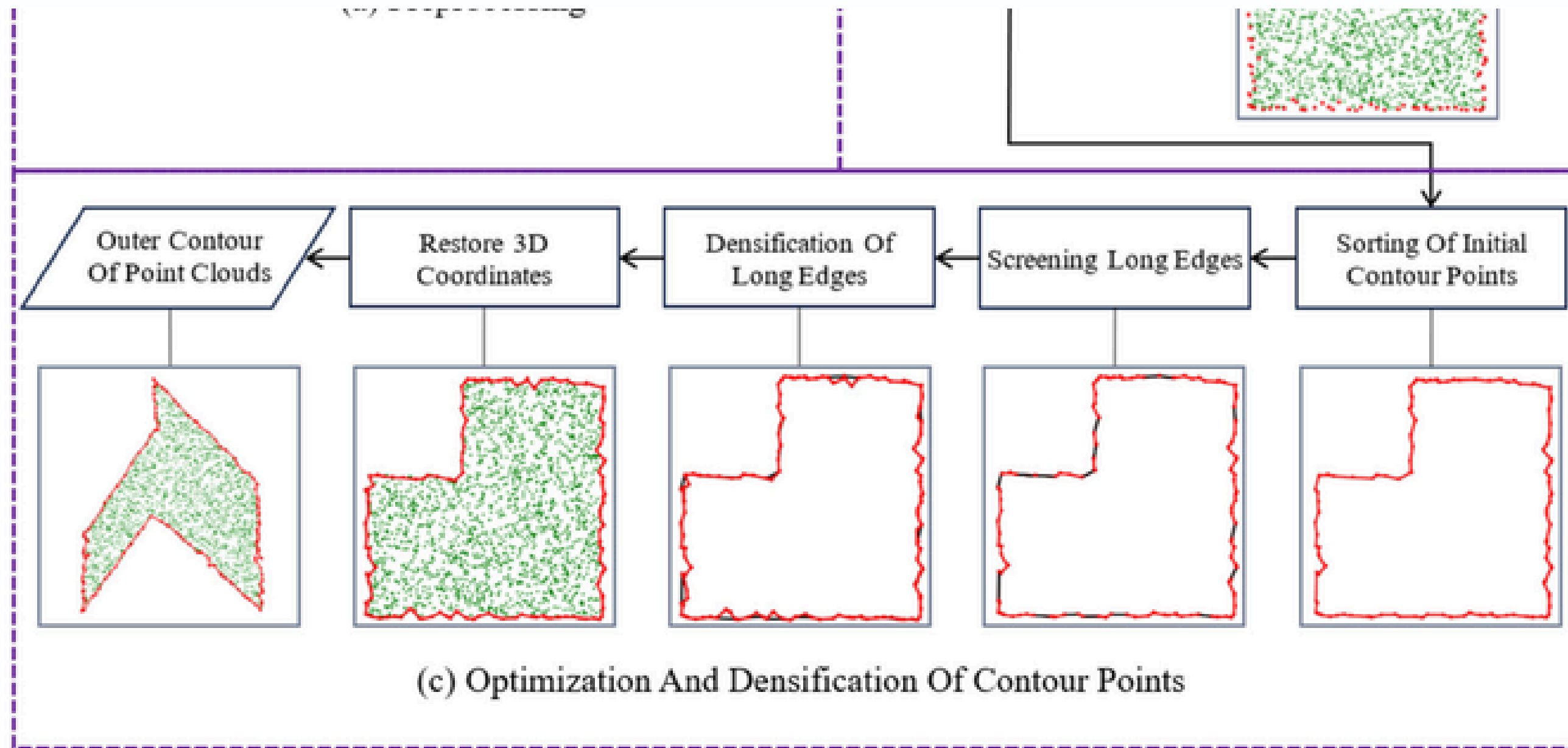
About Proposed Method (Contour Points Extraction - Single Direction)



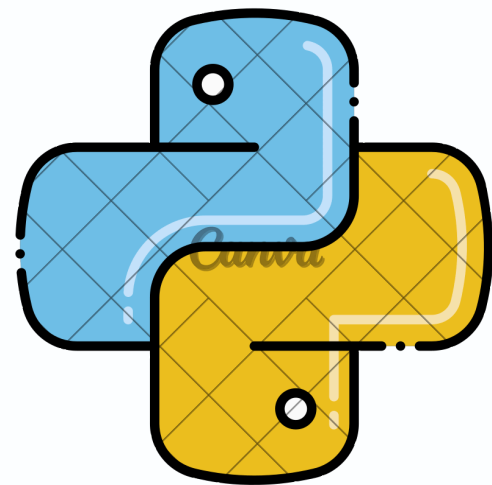
About Proposed Method (Contour Points Extraction - Multidirection)



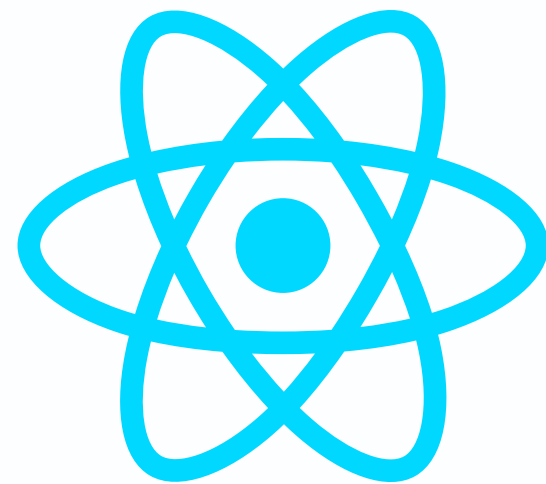
About Proposed Method (Optimization and Densification)



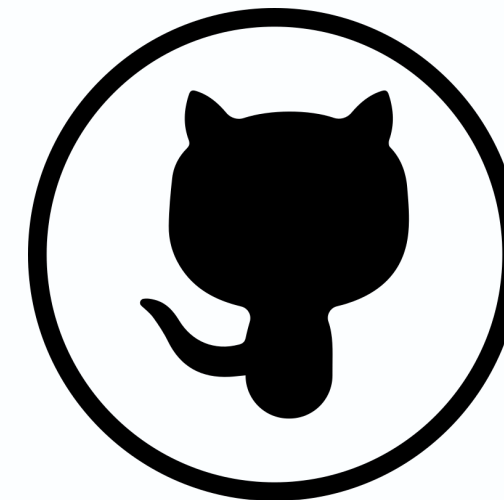
Tools



Python



Reactjs



Github

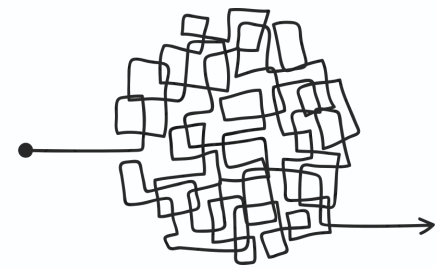
Project Demonstration



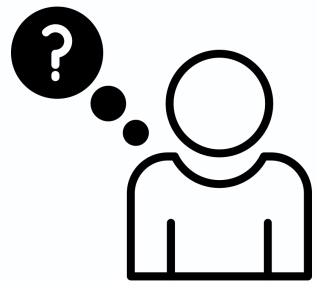
App

Test Case & Results

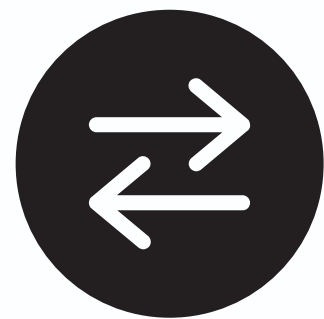
TC002 2D Contour Visualization	
Test Scenario	Ensure that the extracted contour is correctly visualized over the original point cloud in 2D view.
Input / Precondition	Any uploaded and processed point cloud with at least one detected building.
Steps	1. Upload a point cloud and process it. 2. Switch between "Points Only" and "With Contour" view modes. 3. Verify original points and contour are displayed together in overlay mode. 4. Confirm the stats box shows: total points, contour point count, and area. 5. Switch to 3D View and back to 2D View. 6. Verify all points and contour remain visible after switching back (no blank canvas).
Expected Outcomes	• Original points rendered in blue (overlay mode) or darker colour (points-only mode). • Extracted contour rendered as a red line with red vertex dots. • The stats box in the top-left corner shows Points, Contour pts, and Area correctly. • After switching from 3D to 2D and back, the canvas re-renders correctly. • No blank canvas or missing points after tab switching.
Result	Passed - 2D visualization renders correctly including after 3D/2D tab switching.



Contour Sorting on Complex Shapes



Sensitivity of Average Point Spacing Estimation to Outliers



2D/3D View Switching



Multi-Building Batch Processing



Roof Type Classification((flat, gabled, hipped)

References

[1] <https://www.mdpi.com/2072-4292/16/1/190>

[2] <https://ieeexplore.ieee.org/document/10585281>

<https://github.com/RejaulBSSE1324/SPL3>

[Documentation](#)

THANK YOU

